MIL-L-2574B
18 June 1975
SUPERSEDING
MIL-L-2574A
31 May 1955

MILITARY SPECIFICATION

LENS SET, TRIAL, OPHTHALMIC, LARGE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of large trial lens set suitable for ocular refraction.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-621	Box, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Shipping, Fiberboard

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

FSC 6515

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3. REQUIREMENTS

3.1 Material.

3.1.1 Lenses. The lenses shall be high quality ophthalmic crown glass with an index of refraction of 1.5230 ± 0.0015. The lenses shall be additive vertex power type.

3.1.2 Rings and handles. Unless otherwise specified, rings and integral handles shall be fabricated of aluminum . The metal shall have a durable nonreflecting black finish, or be anodized and color coded for fast identification.

3.1.3 Case and tray. Case and tray shall be wood or plastic. When wood is used it shall be oak or mahogany, kiln dried, well-seasoned and free from knots, windshakes, or other defects. When plastic is used the plastic shall be a suitable impact resistant acrylonitrile-butadiene-styrene or high impact resistant molded styrene. The case shall be reinforced with brushed aluminum at the mating edge in the exterior. As an alternate case shall be covered with a durable vinyl simulated leather covering with or without aluminum reinforcements.

3.2 Description. The trial lens set shall consist of a case with hinged cover and removable tray containing components as specified herein.

3.2.1 Test lenses. Each test lens shall be circular and shall be mounted in a ring approximately 1-1/2 inches in diameter. Except for the group B prism lenses (see 3.2.2.3), each ring shall be provided with an integral handle approximately 1/2 inch long. Group B prism lenses shall be mounted in rings as described above, with or without handles. The mounting shall be stamped or engraved with dioptric markings in such a position that the markings shall be visible when the mounted lenses are in their proper position within the tray. The handle shall be perforated with a plus sign for convex lenses and a circle for concave lenses. As an alternate, the ring and handle mountings shall have plus or minus markings and be colored black for plus spheres, red for minus spheres and blue for prisms and accessories. Rings of cylinder lenses shall be permanently marked with axis lines which shall be accurately aligned with the axis of the cylinder lens and shall extend onto the lens (in corresponding positions) for a distance of approximately 1/32 inch. Rings of prism lenses shall be marked with base-apex lines which shall be accurately aligned with the base and apex of the prism lens, the marking to include a small triangle indicating the location of the prism base. The base line of prism lens mounts shall extend onto the base of the prism lens for a distance of approximately 1/32 inch. All markings of the lens mounts shall be in a color contrasting with the color of the mount and shall be legible and clearly cut.

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3.2.2 Lens components. Lens components of the set shall be the following trial lenses.

3.2.2.1 Spheres. 31 pairs plus and 31 pairs minus in the following diopter powers:

0.12	3.50
0.25	3.75
0.37	4.00
0.50	4.50
0.62	5.00
0.75	5.50
1.00	6.00
1.25	6.50
1.50	7.00
1.75	7.50
2.00	8.00
2.25	9.00
2.50	10.00
2.75	11.00
3.00	12.00
3.25	
One pair plus	One each plus and one each minus
8.50	13.00
9.50	14.00
10.50	16.00
11.50	18.00
12.50	20.00
One plus	
15.00	

3.2.2.1.1 Tolerances. Tolerances shall be as follows:

0 to 10.00D $\pm$ 0.04D
10.00 to 15.00D $\pm$ 0.06D
over 15.00D $\pm$ 0.09D

3.2.2.2 Cylinders. 24 pairs plus and 24 pairs minus in the following diopter powers:

0.12	2.50
0.25	2.75
0.37	3.00
0.50	3.25
0.62	3.50
0.75	4.00
1.00	4.50
1.25	5.00
1.50	5.50
1.75	6.00
2.00	7.00
2.25	8.00

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3.2.2.2.1 Tolerances. Tolerances shall be as follows:

0 to 3.75D ± 0.04D
4.00 to 8.00D ± 0.06D

3.2.2.3 Prisms. 17 total as follows:

Group A: 1 pair each:

0.50
1.00
1.50
2.00
3.00
4.00
5.00

Group B: One each

6.00
8.00
10.00

3.2.3 Accessories. The following accessories shall be furnished:

(1) Red Multiple Maddox Rod.
(2) White Multiple Maddox Rod.
(3) Blinder.
(4) Red Glass.
(5) Pinhole Disc.
(6) Stenopaic Disk
(7) Three polarized lenses.
(8) Split prism.

The accessories shall be mounted in circular rings with integral handle as described in 3.1.2. The handle of the polarized lens shall be attached to each ring at an oblique angle to the effective meridian of polarization of the lens. Each lens shall be marked with its effective meridian of polarization by means of a line extending approximately 1/32 inch inward from the ring.

3.2.3.1 Cross cylinders. There shall be two cross cylinder lenses, as follows:

One 0.25 diopter plus and minus lens and
One 0.50 diopter plus and minus lens.

The cross cylinder lenses shall be flat type lenses mounted in a ring with handle. The ring shall be approximately 1-1/2 inches in diameter. Ring and handle shall be fabricated of the material specified in 3.1.2 or nickel plated base metal of nickel silver or brass. Both ends of each cylinder axis of each lens shall be marked with a line terminating in a dot at the end of the marking, toward the center of the lens. The plus cylinder axis shall be marked in white; the minus cylinder axis shall be marked in red. The dioptric powers of each cylinder shall appear adjacent to the cylinder axis marking opposite the handle. The cylinder axes shall be inclined 90 degrees to each

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other and 45 degrees to the vertical. A vertical line approximately 1/8 inch long extending inward from the ring shall be engraved on the lens directly opposite the handle and shall mark the vertical to which the axes are inclined 45 degrees.

3.2.4 Case. The case shall have the following approximate outside dimensions: 21-1/2 inches long, 13-1/4 inches wide and 3-1/4 inches deep. If wood, front, back and sides shall be of 1/2 thickness stock joined smoothly and securely at the corners and mating surfaces. The wooden case shall be covered with imitation leather vinyl to provide an impact, scuff resistant covering. If made of plastic it shall be acrylonitrile butadiene styrene, impact and scuff resistant. Cover shall be recessed inside to maintain clearance of contents when closed but also provide means of lenses staying in place when case is closed and being carried. If necessary, durable plastic foam shall be utilized for this purpose. The cover shall be attached to the case body by two substantial hinges. Brackets shall be provided so that the cover will remain open at approximately 90° when desired. There shall also be two latches for positive closure. Color of the case shall be gray or black.

3.2.4.1 Tray. The tray shall have the following approximate dimensions: 20-1/2 inches long, 12-9/16 inches wide and 1-9/16 inches deep and shall fit inside the case with necessary room for closure. Tray shall be freely removable with no permanent attachments. The tray shall be compartmentalized with separate areas for the various classes of lenses. Each lens compartment shall be properly slotted to hold the lens specified. The compartment slots shall be marked clearly and legibly with the power of the test lens or with letter abbreviations in the case of accessories.

3.2.5 Trial frame. The trial lens frame shall be a revolving 3 cell ophthalmological lens frame with corneal sighting device. Shall have 50 to 80 mm PD adjustments, and extensive range of horizontal and vertical bridge movement. The bridge shall insure comfortable, stable placement on patient's nose. The quickly set temple angle above or below center; and 3-1/2 to 5 inches variable length temples with formed paddle tips shall hold the frame snugly against the patient's head. Shall be capable of holding four trial lenses, a single stationary cell and a triple rotating cell which shall accept any 1-1/2 inch trial lens ring. The trial frame shall be National Stock Number 6515-00-346-2240.

3.3 Identification marking. A metal plate permanently and legibly marked with the item identification and the National Stock Number shall be secured to

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the center of the outside surface of the top of the case. The letters and numbers shall be approximately 1/2 inch in height. A metal plate bearing the name or registered trade-mark of the manufacturer shall be secured in a conspicuous position on to the inside surface of the top of the case. As an alternate, the registered trademark of the manufacturer shall be positioned on the outside surface near the handle of the case.

3.4 Workmanship. Workmanship shall be first class throughout. The lens set (case and components) shall be free from defects which detract from its appearance or may impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, inspection level S-2. Unit of product shall be the case. AQL shall be 1.0 for Major and 4.0 for Minor defects, designated as percent defective.

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4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, inspection level S-1. Acceptance number shall be zero for all sample sizes. Unit of product for test shall be the case.

Table II. Classification of defects

Examine	Defect	Classification	
		Major	Minor
Carrying case	Defective or missing brackets or latches	X	
	Improper design	X	
	Color not as specified or non-uniform		X
	Torn, scratched or cracked		X
Lenses and accessories (including prisms and cross-cylinders)	Lens or accessory missing from set	X	
	Marking improper or missing	X	
	Pits, grayness, scratches, bubbles, striae or inclusions	X	
	Wave distortion	X	
	Clinder curve wrong side		X
	Ring material other than specified		X
	Ring improperly finished or color coded		X
Trial frame	Corneal aligning device missing	X	
	Incorrect design	X	
	PD adjustment not as specified	X	
	Marking or graduations indistinct		X
	Temple length not as specified		X

4.3 Examination. The case shall be examined for defects including, but not limited to, those listed in table.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements, as required.

4.4.1 Measurement of powers. Lens power measurements shall be performed with a standard lens measuring instrument, calibrated to insure accuracy. Tolerances of section 3 shall be defined as focusing tolerances.

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4.5 Examination for the preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of Section.5. Defects shall be scored as specified in Table III. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 defects per hundred units.

Table III Examination for the Preparation for Delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in shipping containers not as specified.
Packaging and packing materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper, location, sequence, or method of application

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit preparation. The lens set shall have the trial lenses, trial frame and accessories placed in their respective compartments in the case. Prior to closing the case, the lenses, frame and accessories shall be secured or cushioned in the case in such a manner so as to prevent movement or damage during shipping and handling.

5.1.2 Level C. Lens sets shall be packaged in accordance with the contractor's commercial practice which will afford the degree of protection required for shipment and handling.

5.2 Packing. Packing shall be level A or C as specified (see 6.2).

5.2.1 Level A. Each set prepared as specified in 5.1.1.1 shall be packaged in a box designed for a type 2 load and constructed in accordance with PPP-B-636, class weather-resistant, variety D.W. Closure, strapping and waterproofing shall be as specified in the appendix of the box specification.

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Note: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level C. Each set shall be packed in a substantial commercial container of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking. Marking shall be as specified in MIL-STD-129.

6. NOTES

6.1 Intended use. The trial lens set is used for ocular refraction and other eye examination procedures.

6.2 Ordering data. Procurement documents should specify the following:
(a) Title, number and date of this specification.
(b) Applicable levels of packaging and packing. (see 5.1 and 5.2).

6.3 This specification is intended to cover the following item appearing in the Federal Supply Catalog:

National Stock Number	Item Identification
6515-00-346-1600	Lens Set, Trial, Ophthalmic, Large

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project Number 6515-1935
is

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-L-2574B LENS SET, TRIAL, OPHTHALMIC, LARGE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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